Sequencing Technology I: Sequencing platforms and chemistry

Introduction to sequencing technology



Goodwin et al. 2016. Nature Reviews Genetics

Next Generation Short Read Sequencing

DNA/RNA Libraries
(A collection of molecules to be sequenced)



Short “reads” of 50 to 300 bp of DNA
from a pool of DNA template molecules

Reads after alignment
to a genome assembly

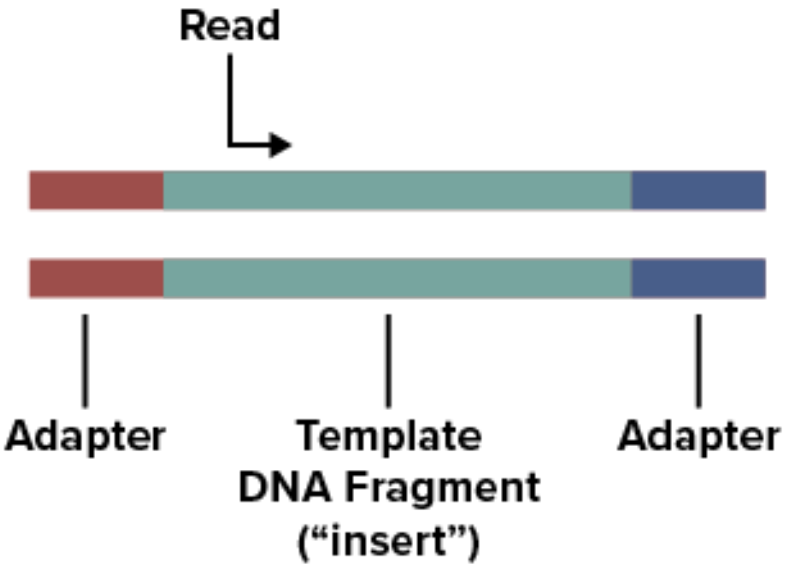


Reference genome assembly

Single-end (SE) sequencing

- Sequencing of a single end of a template DNA molecule, or "fragment"
- Common in RNA-seq applications where read counts are the desired output
- Examples:

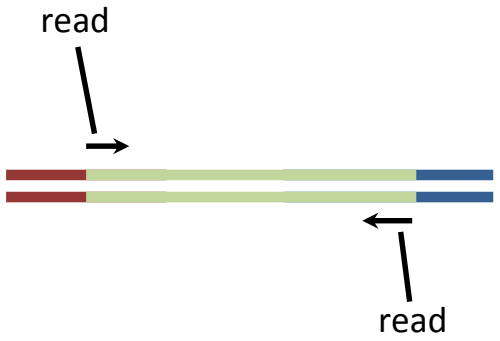
  1 X 50 bp SE
  1 X 100 bp SE
  1 X 150 bp SE



Paired-end (PE) sequencing

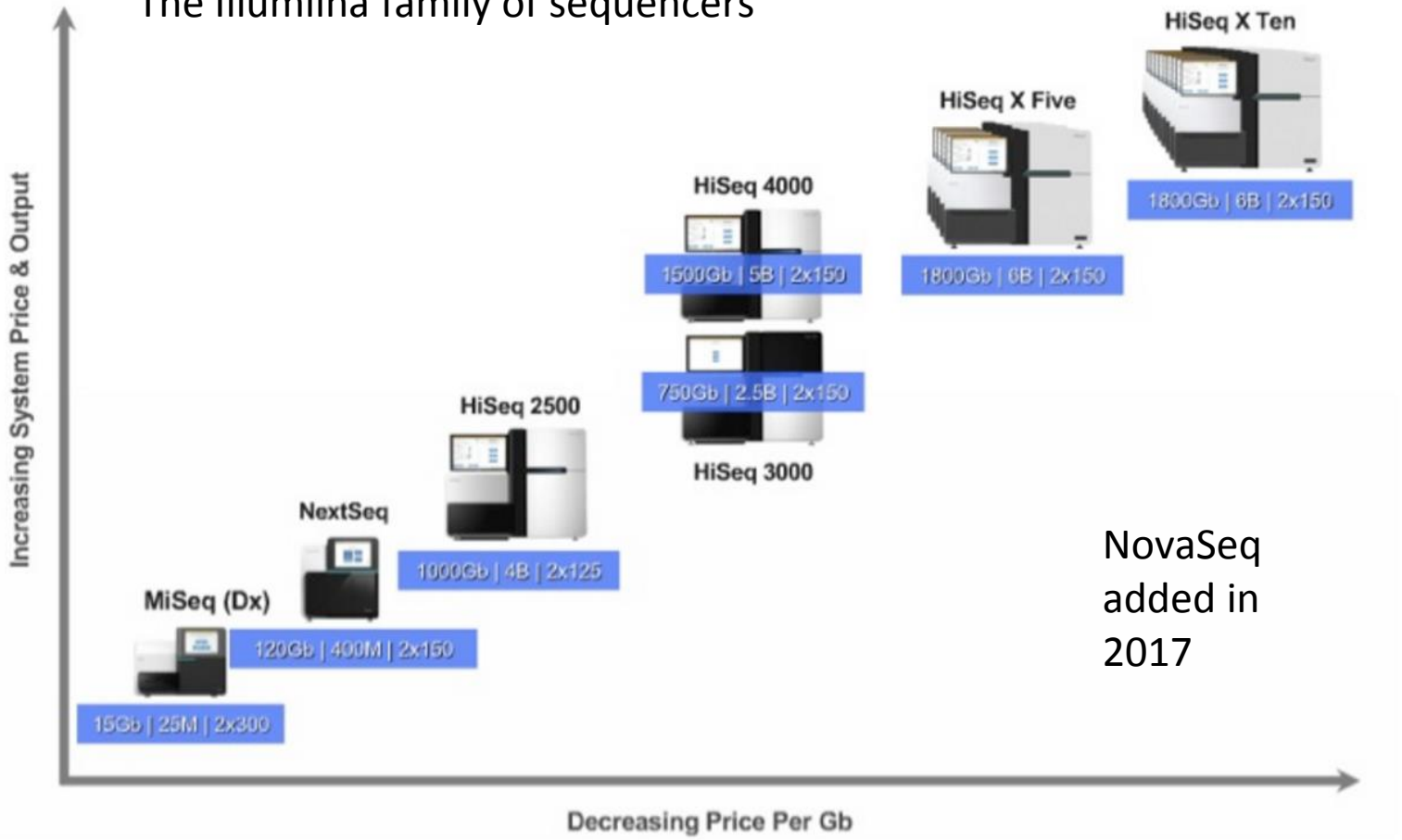
- Sequencing of both ends of a template DNA molecule, or “fragment”
- Library Insert size typically 300-500 bp
- Advantages over single end sequencing
    - alignment
    - *de novo* transcriptome assembly
    - structural variant discovery
    - identification of PCR duplicates
- Mapping orientation is forward-reverse (3’ ends point inward or “innie”) for most short read applications
- Examples:

    2 X 50 PE,
    2 X 100 PE
    2 X 150 PE



Sequencing technologies: Illumina

The Illumiina family of sequencers

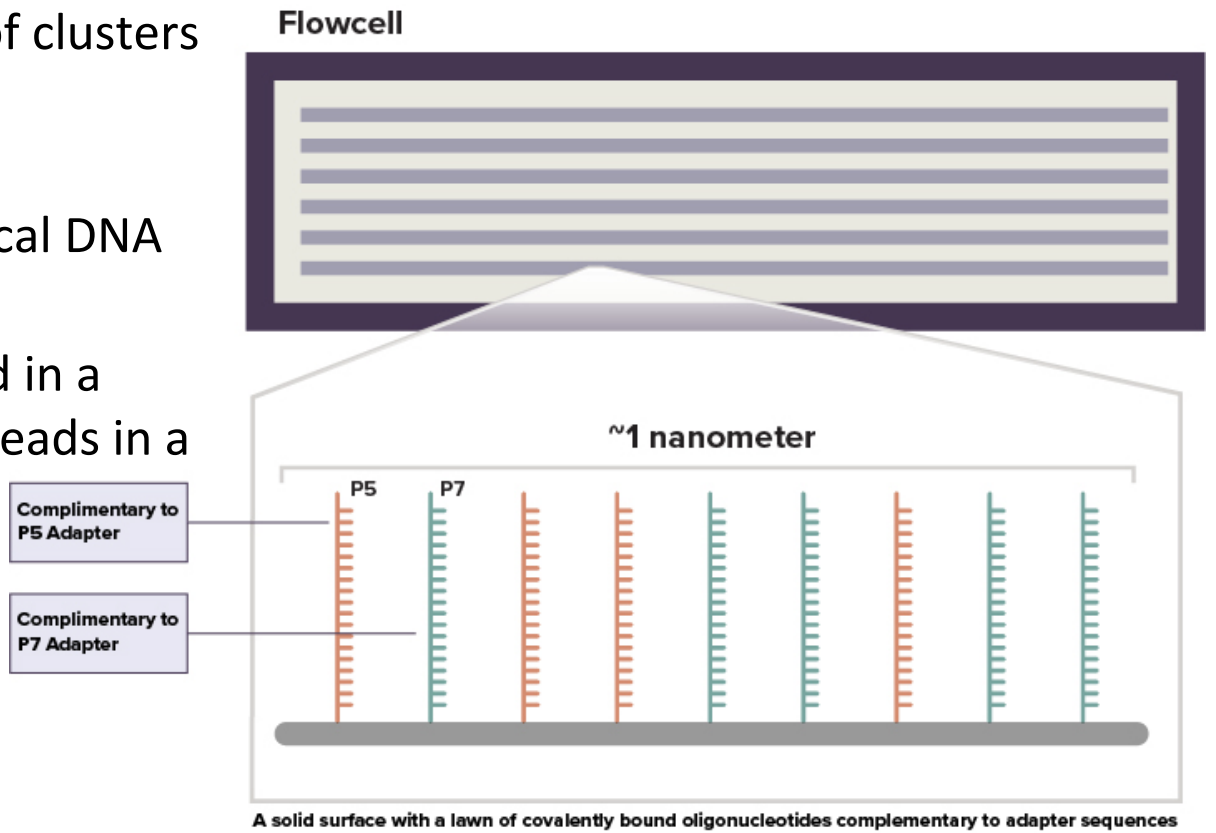


NovaSeq added in 2017

http://blog.genohub.com/illuminas-latest-release-hiseq-3000-4000-nextseq-550-and-hiseq-x5/

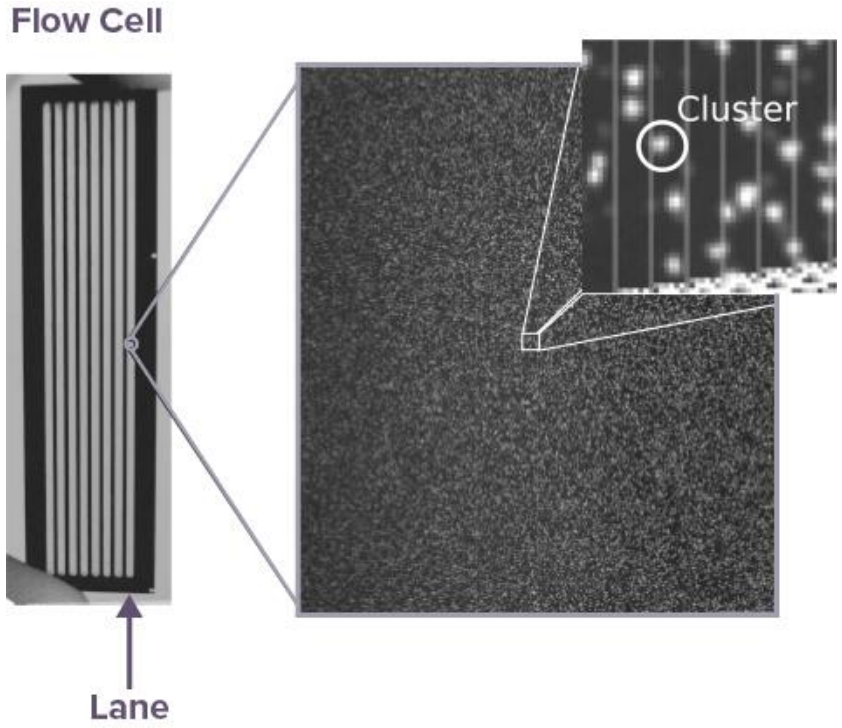
Illumina flow cell

- A solid surface with a lawn of covalently bound oligonucleotides complementary to adapter sequences
- Prior to sequencing, surface-bound oligos seed the formation of clusters of template DNA
- each cluster consists of approximately 1,000 identical DNA template sequences
- Each cluster yields one read in a single-end run or a pair of reads in a paired in run



Illumina flow cell design

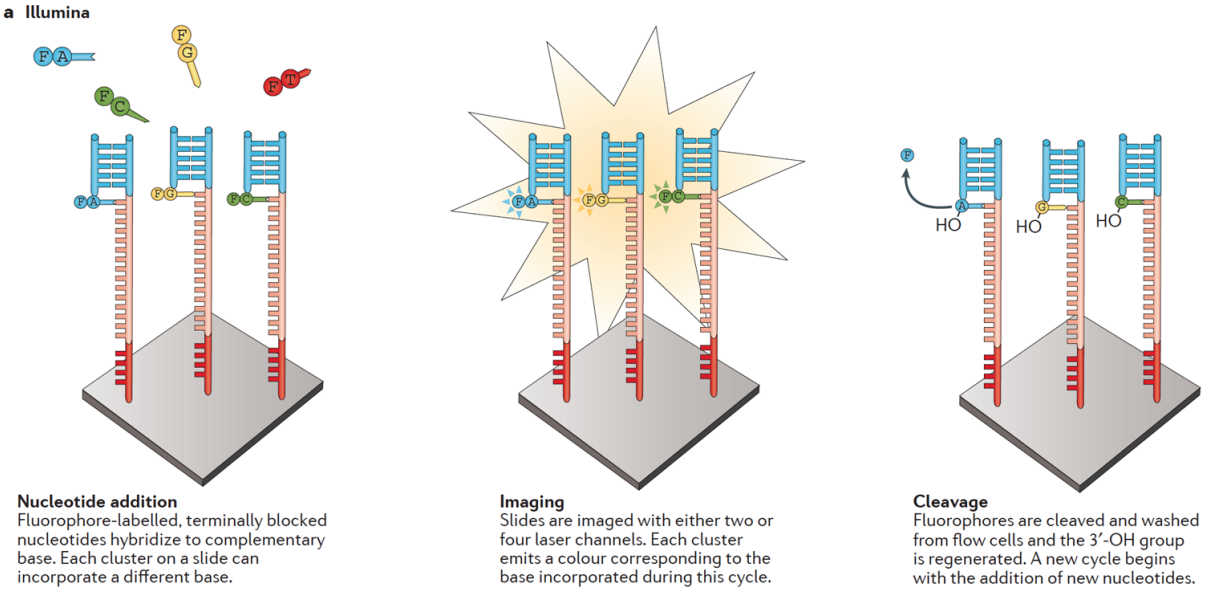
- A “lane” is a unit of a flow cell
- Physically separated from other such units by a physical barrier
- 8 lanes per flow cell on HiSeq family sequencers
- Without barcoding, only one library can be sequenced per lane
- With barcoding (“multiplexing”), up to 384 libraries in a DNA “pool” can be sequenced per lane



Whiteford et al. 2009, Bioinformatics

Illumina: Sequencing-by-synthesis

- “stepwise primer elongation”
- Single-stranded template molecules are primed by a single-stranded oligonucleotide primer
- Reverse terminators are deoxy nucleotide triphosphates (dNTPs, N= A,T,G,or C) with different fluorescent label and a blocked 3’-OH
- The blocked 3’-OH ensures 1 nucleotide base is added per cycle per template molecule
- Each cycle the imager captures fluorescent signals emitted on the flow cell surface and records the signal associated with each cluster
- Cleavage occurs exposing 3’-OH so molecule can be extended in subsequent cycle



Nucleotide addition
Fluorophore-labelled, terminally blocked nucleotides hybridize to complementary base. Each cluster on a slide can incorporate a different base.

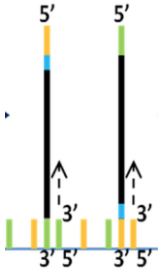
Imaging
Slides are imaged with either two or four laser channels. Each cluster emits a colour corresponding to the base incorporated during this cycle.

Cleavage
Fluorophores are cleaved and washed from flow cells and the 3’-OH group is regenerated. A new cycle begins with the addition of new nucleotides.

Illumina: Sequencing-by-synthesis



(1) Paired-end library with P5 and P7 adapters



(2) Complimentary base pairing of adapter sequences with flow cell-bound oligos.

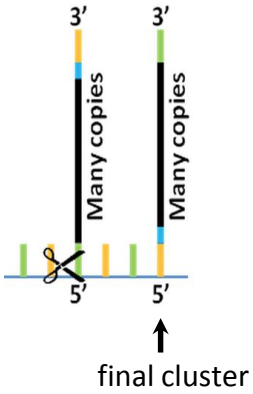
Oligos prime a 5’ to 3’ extension reaction which creates a flowcell-bound complement of the original template.



(3) The original strand is washed away, leaving behind fragment copies that are covalently bound at their 5’ ends to the flowcell surface in a mixture of orientations.

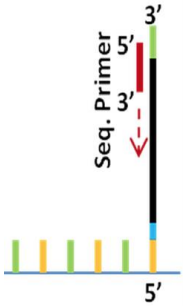
Courtesy: http://nextgen.mgh.harvard.edu/IlluminaChemistry.html

Illumina: Sequencing-by-synthesis

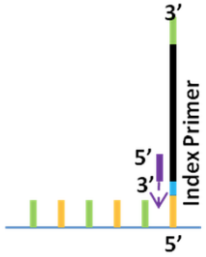


(4) Bridge PCR creates “Many copies” (~1000) of the template in both orientations per cluster

Fragments attached by the P5 adapter are cleaved, resulting in all clusters having P7-bound adapters.



(5) Initiation of the first cycle of sequencing. The P5 sequencing primer anneals to the P5 end of each template and extension proceeds from 5’ to 3’ to create Read 1

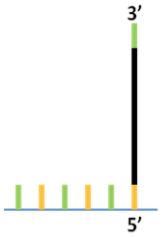


(6) When read 1 is complete the sequenced product is denatured and washed from the flowcell.

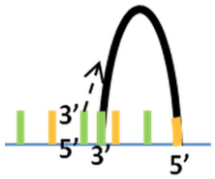
An index primer is added which anneals at the P7 end and primes the sequencing of the barcode, or “index”.

Courtesy: http://nextgen.mgh.harvard.edu/IlluminaChemistry.html

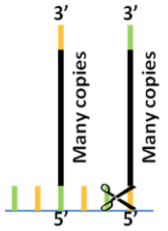
Illumina: Sequencing-by-synthesis



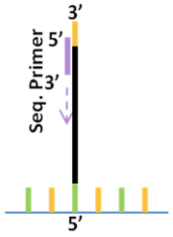
(7) Barcode products are washedfrom the template (P7 adapter is bound to surface).



(8) Bridge amplification is performed to reverse the orientation. This generates both P5 and P7-bound templates



(9) Now P7-bound fragments are cleaved, resulting in templates containing only fragments which are attached by the P5 region.



(10) The P7 sequencing primer is used to initiate extension and sequencing of Read 2

Courtesy: http://nextgen.mgh.harvard.edu/IlluminaChemistry.html

Sequencing-by-synthesis: Data processing

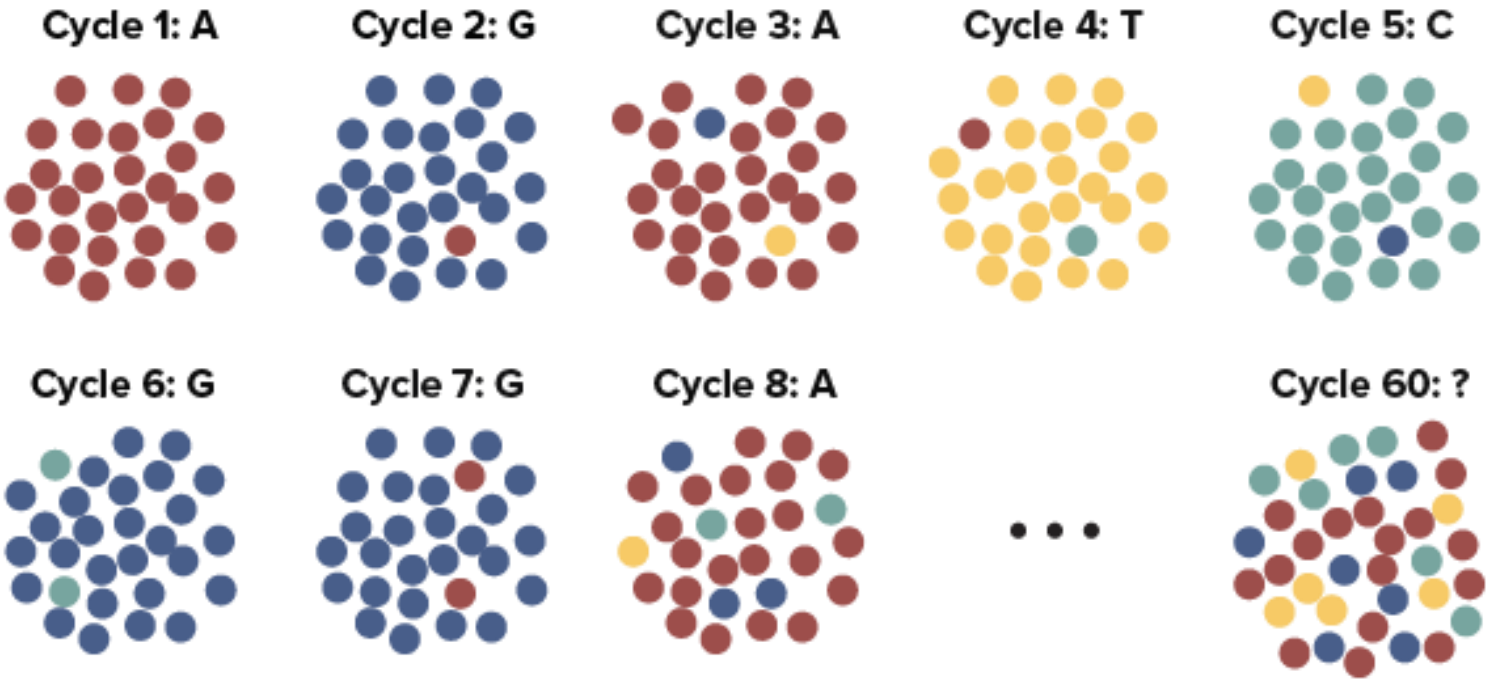
- Illumina instruments generate per-cycle image files which are converted to BCL basecall files as primary outputs

- Illumina's bcl2fastq program translates per-cycle BCL files to fastq

- At the time of conversion, bcl2fastq also demultiplexes the reads and returns separate fastq files for each sample(2 files for PE runs)

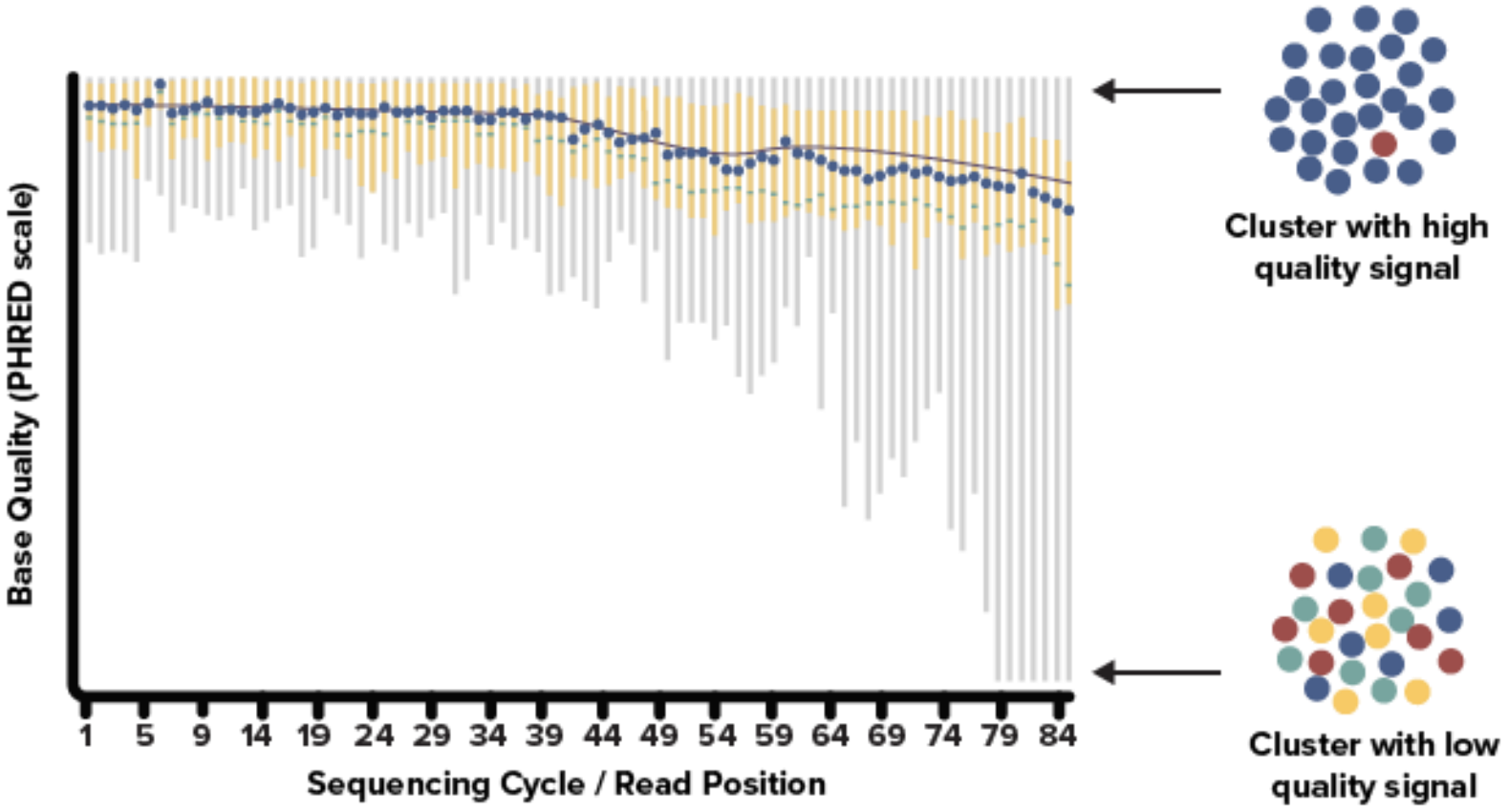
The “dephasing” problem

- Reversible terminators prevent more than one nucleotide addition per cycle.
- Terminators are cleaved from incorporated nucleotides before next cycle
- Cleavage of reversible terminators is imperfect
- If terminator is not cleaved from a template then the signal will be out of phase with other templates in the same cluster (i.e. “dephasing”)



The “dephasing” problem

- Dephasing results in a decline in base quality with increasing cycle



Patterned flow cells

- Newer models of Illumina machines (HiSeq 3000/4000/X, HiSeq X Ten and NovaSeq) use patterned flow cells

- Nanowells on flow cell surface define cluster boundaries

- Single fragments added per well to seed cluster formation (probabilistic based on diffusion and other factors)

- Fragments are amplified to formed clusters

- Proprietary ExAmp chemistry can cause “index swapping” / “spreading-of-signal” problem in multi-plexed experiments

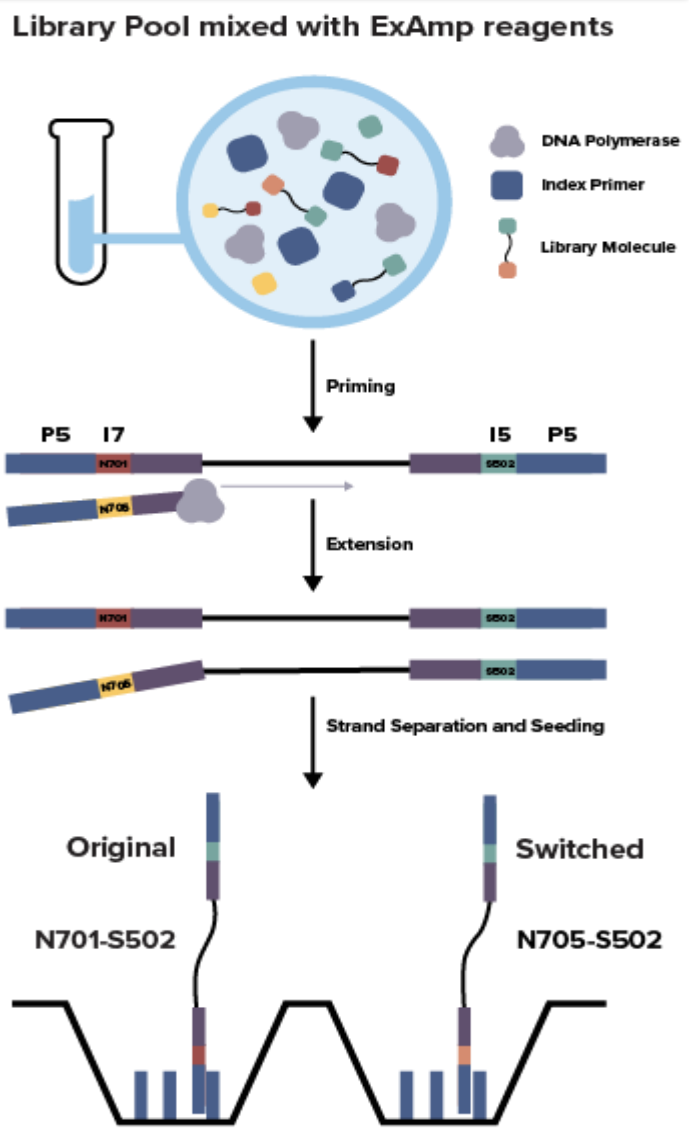
- indexes are swapped before sequencing leading to fragment being assigned to wrong sample when de-multiplexed

https://www.molecularecologist.com/2017/04/right-reads-wrong-index-concerns-with-data-from-illuminas-hiseq-4000/

Hypothesized cause of index swapping on HiSeq 4000

- Swapping hypothesized to occur in ExAmp library before seeding patterned flow cell surface

- Solution: non-reduncdant dual indexing



Sinha et al. BioXRiv (2017)

Benchtop sequencers (e.g., MiSeq)

- End-to-end single benchtop sequencing
- Up to 15 Gb of sequence data per run (15Gb / 3Gb = 5-fold coverage of human genome)
- Applications include rapid diagnostic assays, targeted RNA-seq, tumor profiling, mutagenesis (clone-checking), small genome sequencing, and meta-genomics



Example: Illumina HiSeq 2500

- 2 flow cells per machine
- 8 lanes per flow cell
- 400 Gb bp per flow cell

  400 Gb / 3 Gb = 133 fold-coverage of human genome
- 250,000,000 reads per lane
- Sequencing error rate ~0.1%
- Run time for paired-end data ranges from roughly 5-11 days



Example: NovaSeq 6000

- Patterned flow cells
- Sequencing pools can be run on one (S1), two (S2) or four lanes (S4)
- Up to 3000 Gb of sequence per run
- 3000 Gb / 3 Gb = 1,000 fold-coverage of human genome
- S4 flow cell with 2 X 150 bp reagents = ~ $36,000



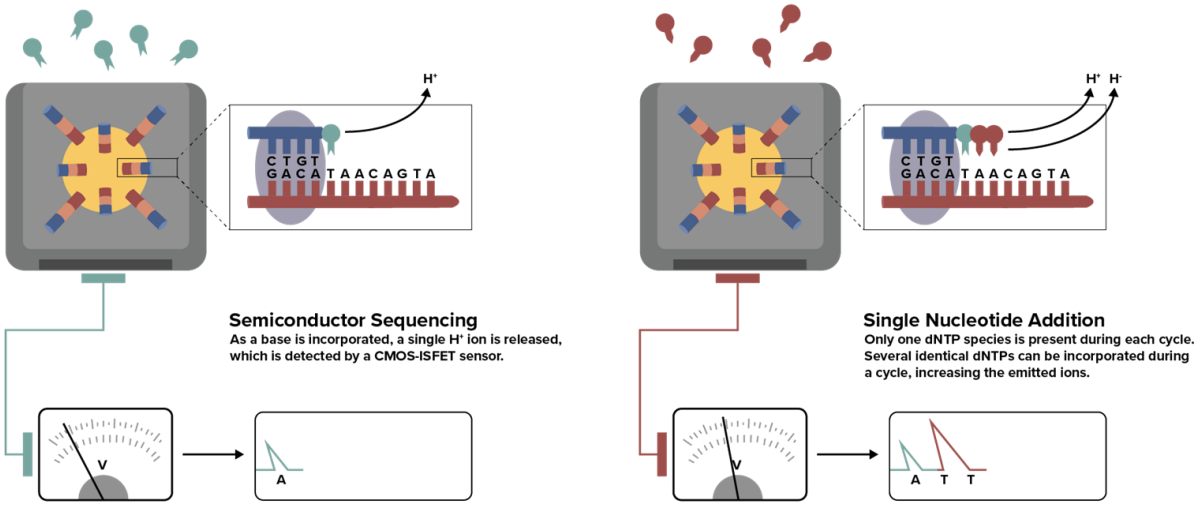
NovaSeq 6000



NovaSeq S2 Flow cell (NYUAD, September 2018)

Sequencing technologies: Ion Torrent

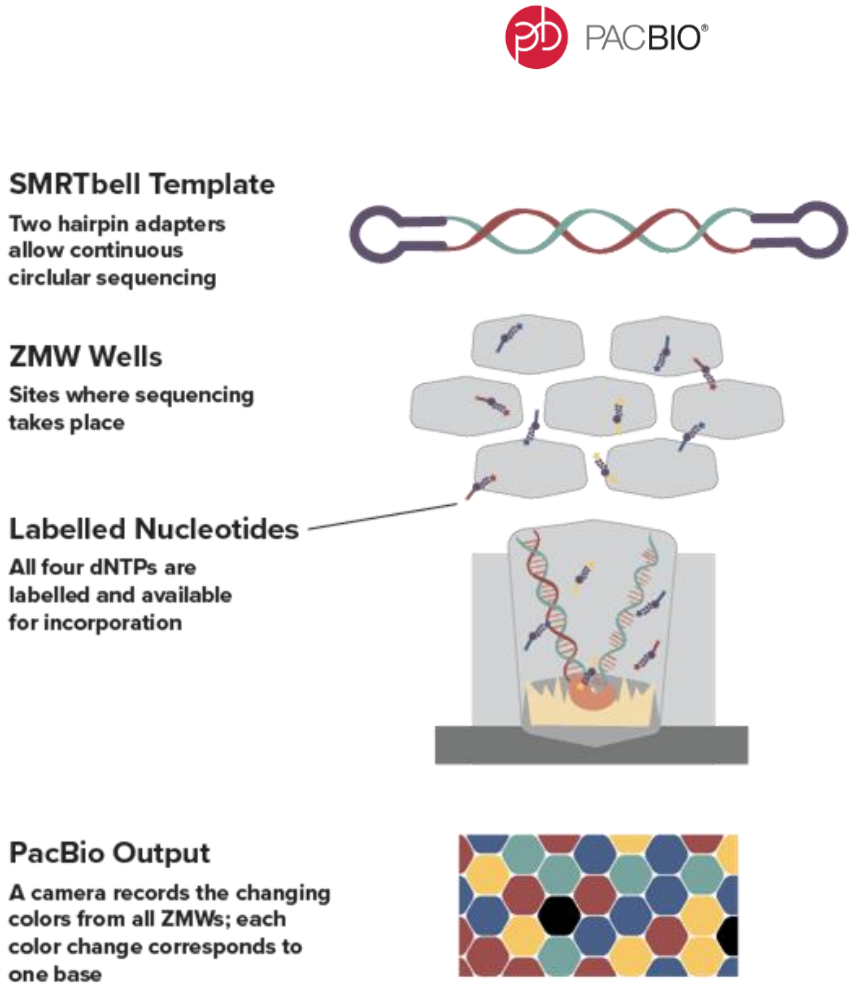
- Short template sequences amplified on beads, or Ion Spheres, 1 template sequence per bead
- Beads are placed in wells (1 bead per well) on chip where sequencing reaction occurs
- Semiconductor chip contains 1.2+ million micro-wells per chip
- Four nucleotides (A,T,G,C) added sequentially
- H^+ ion released when nucleotide is incorporated by DNA polymerase
- Ion sensitive layer detects pH change and converts to voltage
- Signal is translated into DNA sequence separately for each micro-well



Goodwin et al. 2016. Nature Reviews Genetics

PacBio **S**ingle **M**olecule **R**eal **T**ime (SMRT)

- Flow cell consists of thousands of picoliter (microscopic) chambers (zero-mode waveguides, ZMW)
- X million chambers per cell (Sequel)
- DNA polymerase immobilized at the base of the ZMW chamber
- Long DNA template progresses through DNA-pol complex at bottom of ZMW
- Nucleotides (dNTPs = A,T,G,C) are each labelled with a different fluorophore on on terminal phosphate
- At time of incorporation, a nucleotide complementary to the DNA template emits a flourescent signal when terminal phosphate is cleaved
- A laser and camera system records the color and and duration of emitted light
- Each ZMW yields a sequencing read of unspecified length



Goodwin et al. (2016) Nature Reviews Genetics

Pacific Biosciences (PacBio) family of Sequencers

- RSII
- Sequel System



PacBio RSII

Pacific Biosciences (PacBio) family of Sequencers

- Single molecule sequencing
- Read lengths average 10-14 kb (can exceed 100 Kb)
- High error rates (13-15%)
- Applications include whole genome *de novo* assembly, whole transcript sequencing, haplotype sequencing
- PacBio (RSII and Sequel System)



PacBio RSII

Oxford Nanopore sequencing technology

- Flow cell consists of an application-specific integrated circuit (ASIC) chip with ~500 channels (MinION)
- Each channel contains a protein pore complex capable of sensing and passing DNA through the pore
- A voltage differential is applied which introduces a current through the pore
- Different k-mer sequences occupy the pore to a varying extent blocking ion transfer to differing degrees
- each k-mer therefore introduces characteristic effects on the current which is translated into DNA sequence
- Redundancy in sequencing from over-lapping k-mers and bidirectional sequencing of each strand improve sequence accuracy

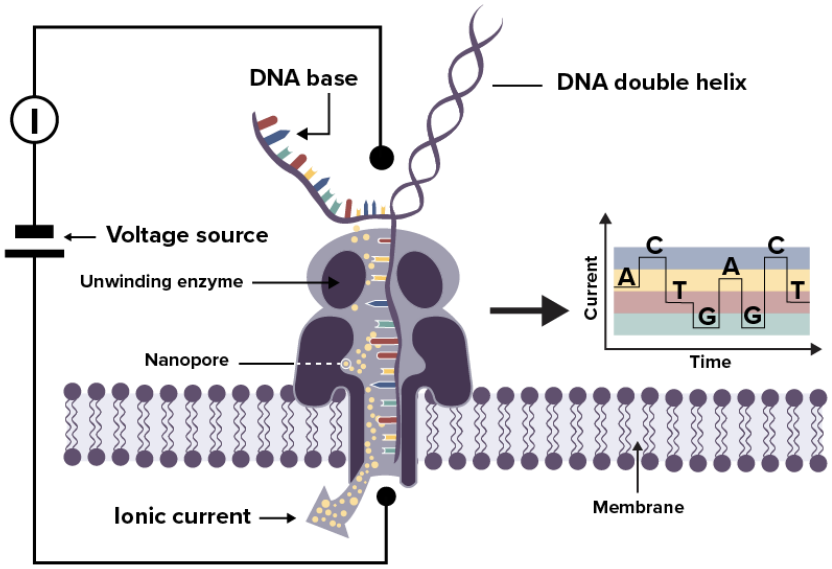


Figure courtesy of Kristin Gopfrich

Oxford Nanopore family of sequencers

Smidg**ION**



PromethION





Min**ION**



Flongle



Grid**ION**

Distinguishing feature(s):

- Portable
- No PCR
- Indeterminate run lengths

https://www.foodnavigator.com/Article/2018/03/21/Oxford-Nanopore-raises-funds-to-support-commercial-expansion

Minor sequencing technologies

- Ion Torrent Personal Genome Machine (PGM) Up to 2 Gb per run

  2 Gb / 3 Gb human genome size = 0.67 fold-coverage per run
- Read lengths of 35-400 bp
- Error rate ~1%
- Small scale applications (microbial genome and transcriptome sequencing, PCR amplicon sequencing)

- Solid 4 (Roche/Applied Biosystems – discontinued 2016)
- 454 (454 Life Sciences – discontinued 2015)



Ion Torrent PGM

https://www.thermofisher.com/order/catalog/product/4462921

Goodwin et al. 2016. Nature Reviews Genetics

Comparison of sequencing technologies

Platform	Error Rate	Read Length	Throughput (maximum)	Cost per Gb
MiSeq v3	0.1%	2 X 75 PE	3.8 Gb	$250
HiSeq 2500	0.1%	2 X 100 PE	400 Gb	$45
PacBio RSII	13%	20 kb	20 Gb	$1000
MinION	? (high)	200 kb	1.5 Gb	$750

After Goodwin et al. 2016. Nature Reviews Genetics